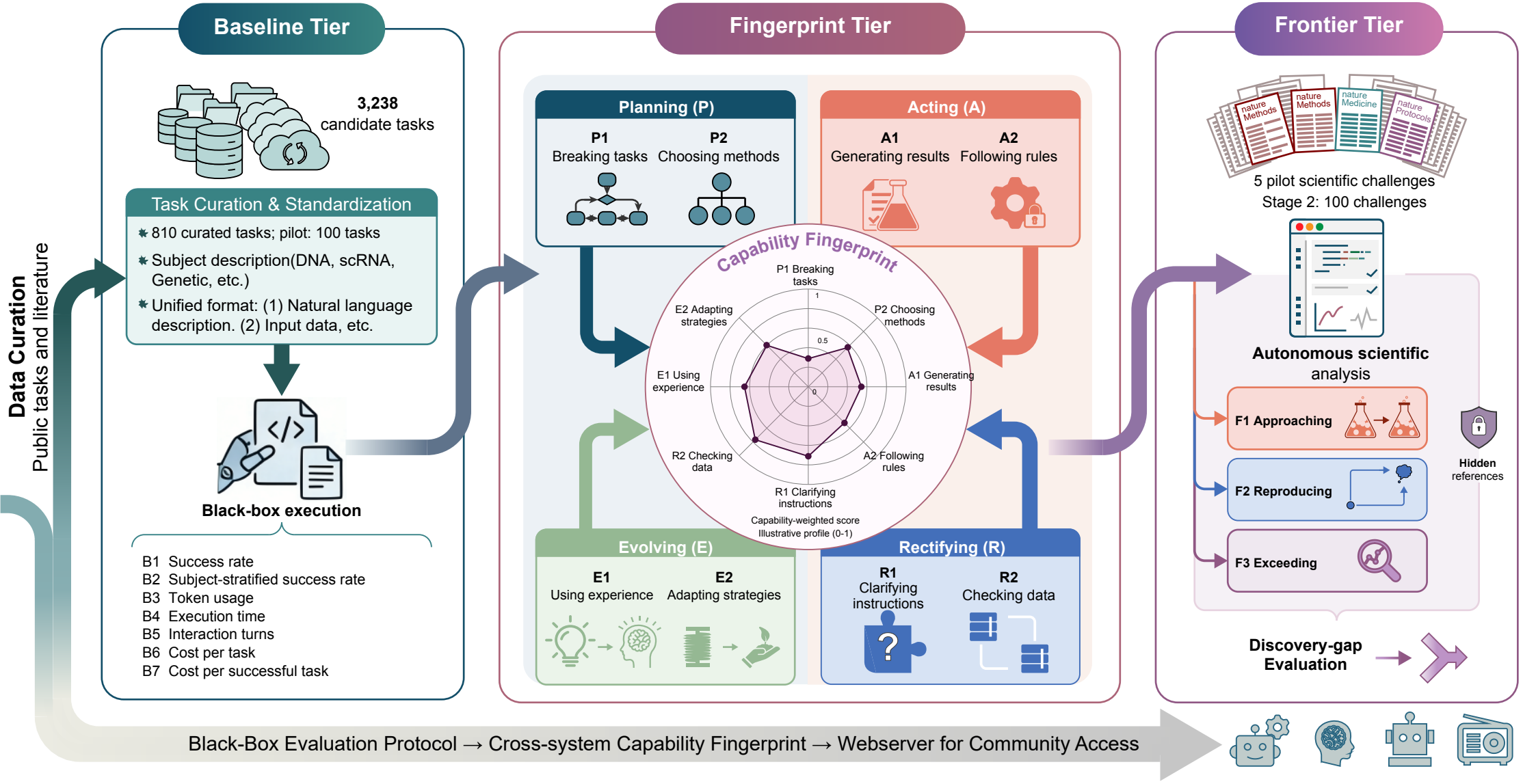

Baseline Tier
Fingerprint Tier
Frontier Tier
Data Curation
Public tasks and literature
3,238
candidate tasks
Task Curation & Standardization
810 curated tasks; pilot: 100 tasks
Subject description(DNA, scRNA, Genetic, etc.)
Unified format: (1) Natural language description. (2) Input data, etc.
Black-box execution
B1 Success rate
B2 Subject-stratified success rate
B3 Token usage
B4 Execution time
B5 Interaction turns
B6 Cost per task
B7 Cost per successful task
Planning (P)
P1 Breaking tasks
P2 Choosing methods
Acting (A)
A1 Generating results
A2 Following rules
Capability Fingerprint
P1 Breaking tasks
P2 Choosing methods
A1 Generating results
A2 Following rules
R1 Clarifying instructions
R2 Checking data
E1 Using experience
E2 Adapting strategies
Capability-weighted score
Illustrative profile (0-1)
Evolving (E)
E1 Using experience
E2 Adapting strategies
Rectifying (R)
R1 Clarifying instructions
R2 Checking data
5 pilot scientific challenges
Stage 2: 100 challenges
Autonomous scientific analysis
F1 Approaching
F2 Reproducing
F3 Exceeding
Hidden references
Discovery-gap Evaluation
Black-Box Evaluation Protocol → Cross-system Capability Fingerprint → Webserver for Community Access